

Design note: state space and the Kalman filter

Status: DRAFT for review. Every DECISION box below is a recommendation awaiting Mico's sign-off; the mathematics in the appendix is to be reviewed equation by equation before any code. Companions: DESIGN_VCE.md, DESIGN_TSINFER.md. Primary references: Durbin & Koopman, *Time Series Analysis by State Space Methods* (2nd ed., 2012) — “DK” below; Koopman (1997); Koopman & Durbin (2000, 2003).

0. What this tier buys

One engine (the Kalman filter/smoothen with exact diffuse initialization and ML estimation) and four front-ends in order of delivery: ucm (unobserved components), exact-ML arima, dfactor (dynamic factors), and a subset of sspace. The engine also opens the staged doors from earlier notes: multivariate lpirf and vec are unrelated, but forecasting with ragged edges, mixed-frequency nowcasting, and EM estimation all become incremental once this exists.

1. Canonical form

The internal representation is DK's:

$$\begin{aligned} y_t &= Z \alpha_t + \epsilon_t, & \epsilon_t &\sim N(0, H) \\ \alpha_{t+1} &= T \alpha_t + R \eta_t, & \eta_t &\sim N(0, Q) \end{aligned}$$

y_t is $p \times 1$, α_t is $m \times 1$, R is $m \times r$. Time-invariant Z, H, T, R, Q only in this tier. H is DIAGONAL (required by univariate filtering, §3; every planned front-end satisfies it). No correlation between ϵ s and η s (rarely used; a documented non-goal).

DECISION 1 (recommended): DK form; time-invariant; H diagonal; uncorrelated disturbances.

2. Initialization

Per-state classification into stationary and diffuse blocks:

- Stationary block: a_1 from the unconditional mean (0 for mean-deleted states), P_{star_1} from the Lyapunov equation $\text{vec}(P) = (I - T(x)T)^{-1} \text{vec}(RQR')$ restricted to the block.
- Diffuse block: EXACT diffuse initialization (Koopman 1997): $P_1 = P_{\text{star}_1} + \kappa P_{\text{inf}_1}$ with $\kappa \rightarrow \infty$ handled analytically by carrying P_{star} and P_{inf} separately through the first d observations until $\text{rank}(P_{\text{inf}})$ collapses to zero.

The large- κ approximation is REJECTED on numerical-constitution grounds: it manufactures $O(\kappa)$ cancellation, which is exactly the arithmetic that diverges across BLAS implementations and would break the byte-identity promise. Exact diffuse is more code and cleaner numbers, and it is what Stata and statsmodels compute, so verification stays exact.

DECISION 2 (recommended): exact diffuse for nonstationary states; Lyapunov unconditional moments for stationary states; no κ .

3. Filter mechanics

UNIVARIATE (sequential) processing, DK §6.4: each element of y_t is treated as a scalar observation updated in turn. Consequences, all favorable:

- No matrix inversion or Cholesky in the update loop — $F_{\{t,i\}}$ is a scalar. Best possible determinism story across BLAS backends.

- Partially missing observation vectors are handled by simply skipping the missing elements — no case analysis, no reindexing. (The WEO panels are full of ragged edges; this is the everyday case, not the corner.)
- Composes cleanly with exact diffuse (Koopman & Durbin 2000): the diffuse recursions are also scalar.

Cost: H must be diagonal (accepted in §1).

DECISION 3 (recommended): univariate filtering throughout; no multivariate covariance filter, no square-root filter.

4. Missing data and the sample contract

Fully missing y_t : prediction step only. Partially missing y_t : update on the observed elements (free under §3). The ts-tier contiguity guard is RELAXED for this tier: the time index must be gap-free after tsset, but missing VALUES inside the range are legal and handled by the filter — that is half the point of the Kalman filter. The estimation output reports the effective number of observations (scalar updates performed).

DECISION 4 (recommended): as above.

5. Smoother

DK backward recursions (§4.4): smoothed states $\alpha_{\hat{t}}$ and variances V_t via (r, N); disturbance smoother for $\epsilon_{\hat{t}}$, $\eta_{\hat{t}}$ and the auxiliary (standardized) residuals. Diffuse-period smoothing via the (r0, r1, N0, N1, N2) extension (DK §5.3), matching KFAS/statsmodels. This buys predict, smstates / predict, rstates / auxiliary-residual diagnostics, and an EM option later, for one backward pass.

DECISION 5 (recommended): full DK smoother incl. disturbance smoother, from day one.

6. Likelihood and estimation

Prediction-error decomposition in scalar form: each non-diffuse scalar update contributes $-0.5 (\log 2 \pi + \log F + v^2 / F)$; each diffuse step with $F_{\text{inf}} > 0$ contributes $-0.5 (\log 2 \pi + \log F_{\text{inf}})$ (Koopman 1997's diffuse likelihood — Stata's and statsmodels' convention; the number of such terms is d, reported).

Optimizer: GSL BFGS2 on the transformed parameters (§7), gradients by central differences (fixed $h = 1e-5$ relative), deterministic line-search constants, convergence when the relative log-likelihood improvement $< 1e-9$ or 200 iterations. Byte-identity stance, stated plainly: an iterative optimizer is the first place where backend rounding can change the PATH, not just the seventh digit. Defenses: (a) tight convergence so all backends land within tolerance of the same optimum; (b) regression-test goldens print estimates through round() at 1e-5 rather than raw display precision; (c) a COMPATIBILITY.md paragraph documenting the stance. If a divergence survives (a)-(b) on the three rigs, the release gate fails and we tighten — the promise outranks the feature.

DECISION 6 (recommended): as above.

7. Parameterization

- Variances: $\sigma^2 = \exp(2 \psi)$ (log-sigma unconstrained).
- Covariance blocks (dfactor, later): Cholesky factors, diagonal log-transformed.

- AR/MA polynomials in the arima front-end: Monahan's partial- autocorrelation transform, enforcing stationarity/invertibility during optimization (Stata's arima does the same).
- Generic sspace-subset: coefficients unconstrained (Stata's behavior); a nonstationary T during evaluation is an error with a clear message, not a crash.

DECISION 7 (recommended): as above.

8. Command surface and delivery order

Fixed-structure front-ends first; generic sspace last. The 5% argument: `ucm gdp, model(lltrend)` is typed a hundred times for every hand-written state equation.

1. `ucm depvar [if] [in], model(MODEL) [seasonal(#)]` MODELS in this release: `ntrend` (level only = white noise around const), `llevel` (local level), `lltrend` (local linear trend), `rwalk`, `rwdrift`. `seasonal(#)`: trigonometric... no — DUMMY seasonal (sum-to-zero), one variance. Cycle: STAGED.
2. `arima depvar, arima(p,d,q) [sarima(P,D,Q,s)]` re-grounded on the engine: exact ML via the state-space form, replacing the current `arima.c` likelihood. The old command surface is preserved; the numbers move to exact ML and the change is documented loudly (this WILL change existing `arima` goldens — it is a correction, like Bug 39).
3. `dfactor`: one common factor, AR(1) factor dynamics, idiosyncratic AR(0) errors first (the nowcasting workhorse).
4. `sspace subset`: state equations with fixed coefficient matrices entered via a constrained syntax to be specified in its own addendum once 1-3 are shipped and the engine is trusted.

DECISION 8 (recommended): order as above; `ucm`'s five models + dummy seasonal define release one of the tier.

9. Standard errors

Default: observed information matrix (OIM) by numerical differentiation of the score at the optimum (Stata's `sspace/ucm` default). `vce(robust)`: OPG-Hessian sandwich through the existing `vce` module (its third act). `vce(oim)` and `vce(opg)` selectable.

DECISION 9 (recommended): OIM default, robust via `vce` module.

10. Verification protocol

- Nile local level (DK's running example): `sigma2_eps = 15099`, `sigma2_eta = 1469.1`, and the book's filtered/smoothed trajectories. The Nile series (100 obs) is EMBEDDED as `sysuse nile` so the literature's own numbers become a regression test.
- Airline `arima(0,1,1)(0,1,1)_12` exact ML against Stata's `arima`.
- `statsmodels`: element-by-element comparison of filtered and smoothed states/variances and the diffuse log-likelihood on both problems, plus a partially-missing multivariate problem for the univariate filter path.
- `tools/stata_check_sspace.do` mirrors the regression tests for the IMF Stata run.

Appendix A: the recursions to be implemented

Notation: for observation t , scalar element i with row Z_i of Z and observation variance h_i (diagonal of H). State prediction a , P ($P = P_{\text{star}}$ during and after the diffuse phase; P_{inf} additionally during it).

A.1 Univariate update, standard phase ($P_{\text{inf}} = 0$), skipping missing $y_{\{t,i\}}$:

```

v   = y_{t,i} - Z_i a
F   = Z_i P Z_i' + h_i
K   = P Z_i' / F
a   <- a + K v
P   <- P - F K K'
ll += -0.5 (log 2 pi + log F + v^2 / F)

```

A.2 Univariate update, diffuse phase ($P_{\text{inf}} \neq 0$):

```

v       = y_{t,i} - Z_i a
F_inf   = Z_i P_inf Z_i'
F_st    = Z_i P_star Z_i' + h_i
M_inf   = P_inf Z_i'
M_st    = P_star Z_i'

```

If $F_{\text{inf}} > 0$: $K0 = M_{\text{inf}} / F_{\text{inf}}$ $a \leftarrow a + K0 v$ $P_{\text{star}} \leftarrow P_{\text{star}} + K0 K0' F_{\text{st}} - K0 M_{\text{st}}' - M_{\text{st}}$
 $K0' P_{\text{inf}} \leftarrow P_{\text{inf}} - K0 M_{\text{inf}}'$ $ll += -0.5 (\log 2 \pi + \log F_{\text{inf}})$ Else ($F_{\text{inf}} = 0$): exactly A.1 with
 $F = F_{\text{st}}$.

The diffuse phase ends at the first (t, i) after which $P_{\text{inf}} = 0$ (numerically: $\max |P_{\text{inf}}| < \text{tol_inf}$, $\text{tol_inf} = 1e-8$ relative to its initial scale); d = number of scalar steps with $F_{\text{inf}} > 0$.

A.3 Transition (once per t , after the last element):

```

a <- T a
P_star <- T P_star T' + R Q R'
P_inf <- T P_inf T'           (diffuse phase only)

```

A.4 Backward smoothing, standard phase (DK 4.4, univariate form, elements in reverse order;
 $L = I - K Z_i$):

```

r <- Z_i' v / F + L' r
N <- Z_i' Z_i / F + L' N L

```

and per time step, before stepping to $t-1$: $r \leftarrow T' r$, $N \leftarrow T' N T$ Smoothed: $\alpha_{\text{hat}_t} = a_t + P_t r_{\{t-1\}}$, $V_t = P_t - P_t N_{\{t-1\}} P_t$.

A.5 Diffuse-period smoothing: the $(r_0, r_1, N_0, N_1, N_2)$ recursions of DK §5.3 / Koopman & Durbin (2003), implemented to numerical equality with statsmodels' KalmanSmoother on the verification problems. (Spelled out in the implementation addendum rather than here; the sign conventions differ across editions and the addendum will state the exact update lines being coded, for review.)

A.6 Disturbance smoother (standard phase):

```

eps_hat_{t,i} = h_i (v / F - K' r_t)
eta_hat_t     = Q R' r_t

```

A.7 Lyapunov initialization for the stationary block: solve $(I - T_s(x) T_s) \text{vec}(P) = \text{vec}(R Q R')_s$ by dense LU on the m_s^2 system ($m_s \leq \sim 20$ for every planned front-end; cost irrelevant).

Appendix B: engine architecture

```

src/kalman.c   the engine: no Stata parsing, no printing.
  SSModel { m, p, r; Z, Hdiag, T, R, Q; a1; Pstar1, Pinf1 }
  ss_filter(model, y, T_obs, out F/v/K per step | ll, d)
  ss_smooth(...)           (states, variances, disturbances)
  ss_loglik(model, y)      (thin wrapper used by the MLE)
  ss_mle(model-template, y, theta-map, options)
src/sspace.c   the front-ends: ucm, arima bridge, dfactor,

```

sspace-subset; each builds an SSModel + theta-map
and calls the engine; all printing lives here.

Regression testing stays command-level (house style); the engine gets its coverage through the front-ends plus the Nile/airline goldens.

Appendix C: explicitly staged out

Time-varying system matrices, correlated disturbances, non-diagonal H, square-root filtering, simulation smoother, EM estimation, mixed-frequency observation equations, ucm cycles, multivariate dfactor generalizations, full sspace syntax. Each is an extension of this design, not a revision of it.